Synoptic Question 4: Medical Ultrasound Imaging

Reading Passage

Read the following article and then answer the questions that follow.

Ultrasound imaging is a non-invasive medical diagnostic technique that uses high-frequency sound waves to create images of internal body structures. An ultrasound probe, called a transducer, sends short pulses of ultrasound into the body. These waves travel through different tissues and are reflected at the boundaries between them. The transducer detects the reflected waves (echoes), and a computer uses the time delay and intensity of these echoes to construct an image.

The amount of reflection that occurs at a boundary depends on the difference in acoustic impedance, Z, between the two tissues. Acoustic impedance is a physical property of a medium and is defined as the product of the density of the medium, ρ, and the speed of sound in that medium, c. That is, $Z = \rho c$.

For an ultrasound wave to pass from the transducer into the body efficiently, a coupling gel is applied to the skin. This gel displaces the air between the transducer and the skin, ensuring that most of the ultrasound energy enters the body rather than being reflected at the skin's surface.

As ultrasound travels through tissue, its intensity decreases with distance. This phenomenon is known as attenuation and is caused by both scattering and absorption of the sound energy. The intensity of a sound wave is also related to the amplitude of the vibrations of the medium.

Data:

The properties of several biological tissues are given in Table 2.

Table 1: Acoustic Properties of Tissues

Tissue	**Density, ρ (kg m^{-3})**	**Speed of Sound, c (m s^{-1})**	**Acoustic Impedance, Z (10^6 kg m^{-2} s^{-1})**
Air	1.2	330	4.0×10^{-4}
Fat	950	1450	1.38
Liver	1060	1570	
Muscle	1070	1580	1.69
Bone	1900		7.80

Questions

(a) Complete Table 2 by calculating:

(i) The acoustic impedance of the liver.

(ii) The speed of sound in bone.

(b) The intensity reflection coefficient, α, at a boundary between two materials is given by:

$$\alpha = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1}\right)^2$$

(i) Calculate the intensity reflection coefficient for an ultrasound wave at the boundary between air and muscle.

(ii) Using this result, explain quantitatively why a coupling gel is essential for effective ultrasound imaging.

(c) An ultrasound pulse travels from the transducer through 5.0 cm of liver and is then reflected at a boundary with a bone.

(i) Using data from Table 2, calculate the total time taken for the pulse to travel to the boundary and return to the transducer.

(ii) The shortest time interval that the transducer can measure between sending a pulse and receiving an echo is 1.0 μs. Calculate the minimum distance that the system can resolve.

(d) The resolution of an ultrasound image depends on the wavelength of the sound waves. An ultrasound transducer emits waves with a frequency of 3.0 MHz.

(i) Calculate the wavelength of these waves as they travel through muscle.

(ii) The smallest detail that can be clearly distinguished is approximately one wavelength. Suggest one advantage and one disadvantage of using a higher frequency (e.g., 10 MHz) ultrasound for medical imaging.

(e) An ultrasound transducer emits a pulse that lasts for 2.0 μs. The initial power of the pulse is 5.0 W.

(i) Calculate the total energy contained in a single pulse.

(ii) If the transducer emits 1000 pulses per second (the pulse repetition frequency), what is the average power output of the transducer?

(f) The intensity of a sound wave is proportional to the square of its amplitude ($I \propto A^2$).

(i) An echo returns from a tissue boundary with an intensity that is 1% of the initial intensity. Determine the ratio of the amplitude of the echo to the initial amplitude of the pulse.

(ii) Explain why echoes returning from deeper structures have a much lower intensity, referencing at least two physical principles.

Solutions

(a) (i) For Liver: $Z = \rho c = (1060)(1570) = 1.66 \times 10^6$ kg m^{-2}s^{-1}.

(ii) For Bone: $c = \frac{Z}{\rho} = \frac{7.80 \times 10^6}{1900} = 4105$ m s^{-1}.

(b) (i) For air ($Z_1 = 4.0 \times 10^{-4} \times 10^6 = 400$) and muscle ($Z_2 = 1.69 \times 10^6$):

$$\alpha = \left(\frac{1.69 \times 10^6 - 400}{1.69 \times 10^6 + 400}\right)^2 \approx 0.999$$

(ii) The reflection coefficient of nearly 1.0 (or 99.9%) means that almost all the ultrasound energy is reflected at the skin's surface and does not enter the body. A coupling gel has an acoustic impedance similar to skin ($Z_{gel} \approx Z_{skin}$), which makes the reflection coefficient close to zero, allowing most of the ultrasound to be transmitted.

(c) (i) Depth $d = 5.0$ cm $= 0.050$ m. Speed of sound in liver, $c = 1570$ m s^{-1}. Total distance travelled $= 2d = 2 \times 0.050 = 0.10$ m. Time taken, $t = \frac{\text{distance}}{\text{speed}} = \frac{0.10}{1570} = 6.37 \times 10^{-5}$ s (or 63.7 μs).

(ii) The distance corresponds to half the travel time. $d_{min} = \frac{c \times t_{min}}{2} = \frac{(1570)(1.0 \times 10^{-6})}{2} = 7.85 \times 10^{-4}$ m (or 0.79 mm).

(d) (i) Frequency $f = 3.0$ MHz $= 3.0 \times 10^6$ Hz. Speed of sound in muscle, $c = 1580$ m s^{-1}. Wavelength, $\lambda = \frac{c}{f} = \frac{1580}{3.0 \times 10^6} = 5.27 \times 10^{-4}$ m (or 0.53 mm).

(ii) **Advantage:** A higher frequency gives a shorter wavelength ($\lambda = c/f$), allowing for better axial resolution and the ability to distinguish smaller details.
Disadvantage: Higher frequency ultrasound is attenuated more strongly by tissues. This reduces its penetration depth, making it unsuitable for imaging deeper organs.

(e) (i) Energy = Power × Time = (5.0 W) × (2.0×10^{-6} s) = 1.0×10^{-5} J.

(ii) The total energy delivered per second is the energy per pulse multiplied by the number of pulses per second. Average Power = (1.0×10^{-5} J/pulse) × (1000 pulses/s) = 0.010 W.

(f) (i) Given $I_{echo} = 0.01 \times I_{initial}$. Since $I \propto A^2$, we have $\frac{I_{echo}}{I_{initial}} = \left(\frac{A_{echo}}{A_{initial}}\right)^2$.

$$\frac{A_{echo}}{A_{initial}} = \sqrt{\frac{I_{echo}}{I_{initial}}} = \sqrt{0.01} = 0.1$$

The ratio of the amplitudes is 0.1.

(ii) Echoes from deeper structures are weaker for two main reasons:

- **Attenuation:** As the pulse travels deeper into the body, its intensity is reduced due to absorption and scattering by the tissues. This happens on the way to the boundary and on the way back, weakening the echo.
- **Reflection at Interfaces:** At each tissue boundary, only a fraction of the wave's intensity is reflected. The rest is transmitted deeper. Therefore, the initial pulse is already weaker when it reaches a deeper boundary, and the resulting echo is a fraction of that already weakened pulse.

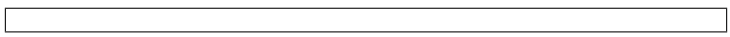

Figure 1: Ultrasound wave interacting with a tissue boundary

Appended CIE Paper 4 Style Materials

This appendix keeps the uploaded paper unchanged and adds CIE-style structured question formats and mark schemes for two variants.

Variant 1 (Core Context, Structured CIE P4 Format)

Question Paper Format

Synoptic Question: Medical Ultrasound Imaging [Total: 30 marks]

Given data (as in the source paper):

Tissue	ρ / kg m^{-3}	c / m s^{-1}	Z / 10^6 kg m^{-2} s^{-1}
Air	1.2	330	4.0×10^{-4}
Fat	950	1450	1.38
Liver	1060	1570	?
Muscle	1070	1580	1.69
Bone	1900	?	7.80

(a) Calculate:

(i) acoustic impedance of liver,

(ii) speed of sound in bone. [4]

(b) The intensity reflection coefficient is

$$\alpha = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1}\right)^2.$$

(i) Determine α for air–muscle boundary.

(ii) Explain, with numerical support, why coupling gel is essential. [4]

(c) A pulse travels through 5.0 cm of liver and reflects from liver–bone boundary.

(i) Calculate round-trip time.

(ii) The minimum measurable interval is 1.0 μs. Calculate minimum resolvable distance. [4]

(d) A transducer emits ultrasound of frequency 3.0 MHz in muscle.

(i) Calculate wavelength.

(ii) State one advantage and one disadvantage of using 10 MHz instead. [4]

(e) A pulse lasts 2.0 μs and has peak power 5.0 W.

(i) Calculate energy per pulse.

(ii) If PRF = 1000 s^{-1}, determine average power. [4]

(f) $I \propto A^2$.

(i) If echo intensity is 1% of initial intensity, find amplitude ratio.

(ii) Explain why deeper echoes are weaker, using at least two physical principles. [10]

Mark Scheme (CIE P4 Style)

Guidance: award marks for correct physics and valid alternative methods.

Part	Marking points	Marks
(a)(i)	$Z = \rho c = (1060)(1570) = 1.66 \times 10^6$ kg m^{-2} s^{-1} (allow 1.66–1.67 $\times 10^6$)	2
(a)(ii)	$c = Z/\rho = (7.80 \times 10^6)/1900 = 4.11 \times 10^3$ m s^{-1}	2
(b)(i)	Correct substitution into α formula with consistent units for Z_{air} and Z_{muscle}; $\alpha \approx 0.999$	2
(b)(ii)	Quantitative statement: nearly all intensity reflected without gel; gel reduces mismatch and increases transmission	2
(c)(i)	Uses round-trip distance $2d = 0.10$ m; $t = 0.10/1570 = 6.37 \times 10^{-5}$ s (63.7 μs)	2
(c)(ii)	$d_{\text{min}} = ct/2 = (1570)(1.0 \times 10^{-6})/2 = 7.85 \times 10^{-4}$ m (0.79 mm)	2
(d)(i)	$\lambda = c/f = 1580/(3.0 \times 10^6) = 5.27 \times 10^{-4}$ m (0.53 mm)	2
(d)(ii)	Advantage: improved resolution due to shorter wavelength; disadvantage: greater attenuation / reduced penetration depth	2
(e)(i)	$E = Pt = (5.0)(2.0 \times 10^{-6}) = 1.0 \times 10^{-5}$ J	2
(e)(ii)	$P_{\text{avg}} = E \times \text{PRF} = (1.0 \times 10^{-5})(1000) = 1.0 \times 10^{-2}$ W	2
(f)(i)	$A_{\text{echo}}/A_0 = \sqrt{I_{\text{echo}}/I_0} = \sqrt{0.01} = 0.10$	2
(f)(ii)	Any four valid points, e.g. attenuation by absorption/scattering, two-way path loss, partial reflection at multiple interfaces, geometric spreading, frequency dependence of attenuation, reduced SNR from noise	8
Total		30

Sample Solution (Variant 1)

Numerical answers:

- (a)(i) 1.66×10^6 kg m^{-2} s^{-1}, (a)(ii) 4.11×10^3 m s^{-1}.
- (b)(i) $\alpha \approx 0.999$, so about 99.9% intensity reflected at air–muscle interface.
- (c)(i) 6.37×10^{-5} s, (c)(ii) 7.85×10^{-4} m.
- (d)(i) 5.27×10^{-4} m, (d)(ii) higher frequency gives better resolution but poorer penetration.
- (e)(i) 1.0×10^{-5} J, (e)(ii) 1.0×10^{-2} W.

- (f)(i) amplitude ratio 0.10.
- (f)(ii) deeper echoes are weaker due to cumulative attenuation and reduced reflected fraction from already weakened transmitted pulses.

Variant 2 (Higher Difficulty, Open-Ended CIE P4 Format)

Question Paper Format

Synoptic Question: Advanced Ultrasound Physics and Imaging [Total: 40 marks]

Use $c_{\text{soft tissue}} = 1540$ m s^{-1} unless otherwise stated.

(a) **Pulse-echo timing and spatial resolution**

(i) A reflector is detected at 78 μs after pulse emission. Calculate depth.

(ii) The timing electronics uncertainty is ± 0.50 μs. Estimate the depth uncertainty.

(iii) Explain how pulse duration limits axial resolution and derive an expression for axial resolution in terms of spatial pulse length (SPL). [7]

(b) **Acoustic impedance matching and transmission design**

(i) Show that pressure-amplitude reflection coefficient is $r = (Z_2 - Z_1)/(Z_2 + Z_1)$ and intensity reflection is $R = r^2$.

(ii) For transducer crystal $Z_c = 30$ MRayl and tissue $Z_t = 1.54$ MRayl, calculate R without matching layer.

(iii) A quarter-wave matching layer is used. Derive the required impedance Z_m and calculate its value.

(iv) Explain physically why quarter-wave thickness improves transmission at one design frequency. [11]

(c) **Attenuation, frequency choice and diagnostic trade-offs** Soft-tissue attenuation coefficient is approximately 0.7 dB cm^{-1} MHz^{-1}.

(i) For depth 8.0 cm, calculate total two-way attenuation at 3.5 MHz and at 10 MHz.

(ii) Hence determine intensity fractions I/I_0 returning from that depth.

(iii) Discuss which frequency is more suitable for (1) deep abdominal imaging and (2) superficial tendon imaging, with quantitative justification. [9]

(d) **Open-ended physics explanation: Doppler ultrasound and limitations** A clinician uses Doppler ultrasound to estimate blood speed in an artery.

(i) Derive or state the Doppler shift relation for reflected ultrasound and define all symbols.

(ii) Explain how beam–flow angle affects measured speed and why angle correction is essential.

(iii) Discuss at least three limitations / error sources in real measurements and suggest improvements. [13]

Mark Scheme (CIE P4 Style)

Guidance: award credit for coherent, logically structured physics arguments in open-ended parts.

Part	Marking points	Marks
(a)(i)	$d = ct/2 = (1540)(78 \times 10^{-6})/2 = 6.01 \times 10^{-2}$ m $= 6.0$ cm	2
(a)(ii)	$\Delta d = c\Delta t/2 = (1540)(0.50 \times 10^{-6})/2 = 3.85 \times 10^{-4}$ m $= 0.39$ mm	2
(a)(iii)	States axial resolution limited by pulse length; derives / states $\Delta x_{\text{axial}} \approx \text{SPL}/2$ with explanation of echo overlap	3
(b)(i)	Correct statement/derivation of r and $R = r^2$	2
(b)(ii)	$R = ((30 - 1.54)/(30 + 1.54))^2 \approx 0.814$ (81.4% reflected)	2
(b)(iii)	Uses $Z_m = \sqrt{Z_c Z_t}$; $Z_m = \sqrt{30 \times 1.54} = 6.80$ MRayl	3
(b)(iv)	Quarter-wave layer gives destructive interference of reflections and constructive transmission at design frequency; bandwidth limitation noted	4
(c)(i)	Two-way path = 16 cm. At 3.5 MHz: $A = 0.7 \times 16 \times 3.5 = 39.2$ dB. At 10 MHz: $A = 112$ dB	3
(c)(ii)	Uses $I/I_0 = 10^{-A/10}$: at 3.5 MHz $\approx 1.2 \times 10^{-4}$, at 10 MHz $\approx 6.3 \times 10^{-12}$	2
(c)(iii)	Quantitative decision: lower frequency for deep imaging, higher for superficial detail; clear trade-off discussion	4
(d)(i)	Correct relation, e.g. $\Delta f = \dfrac{2 f_0 v \cos\theta}{c}$ (or equivalent rearrangement) with symbol definitions	4
(d)(ii)	Explains $v_{\text{meas}} = v \cos\theta$ dependence; error grows as θ increases; need angle estimation/correction	3
(d)(iii)	Any six from: spectral broadening, aliasing, non-uniform flow profile, wall filter effects, poor SNR, incorrect angle, sample-volume placement, attenuation, limited temporal resolution, machine calibration; plus practical mitigations	6
Total		40

Sample Solution (Variant 2)

(a)

- Depth $d = 6.0$ cm.
- Depth uncertainty ± 0.39 mm.
- Axial resolution $\approx$ SPL/2 because two interfaces closer than half pulse length produce overlapping echoes.

(b)

- Without matching layer, $R \approx 0.814$, so most intensity is reflected.

- Optimum single matching layer impedance: $Z_m = \sqrt{Z_c Z_t} = 6.80$ MRayl.
- Quarter-wave thickness causes reflected waves from the two interfaces to be approximately π out of phase at design frequency, reducing net reflection.

(c)

- At 3.5 MHz and 8.0 cm depth: two-way attenuation 39.2 dB, giving $I/I_0 \approx 1.2 \times 10^{-4}$.
- At 10 MHz: 112 dB, giving $I/I_0 \approx 6.3 \times 10^{-12}$.
- Therefore 3.5 MHz is suitable for deep abdominal targets; 10 MHz is better for superficial high-detail imaging.

(d)

- Use $\Delta f = 2 f_0 v \cos\theta / c$.
- Speed estimate requires $v = \Delta f c / (2 f_0 \cos\theta)$, so small angle errors can produce large velocity errors when θ is large.
- Reliable practice includes minimizing beam-flow angle, careful gate placement, adequate PRF to avoid aliasing, repeat measurements, and interpreting traces with flow profile in mind.